MAC addresses

Both addresses in the Ethernet frame are examples of what are called physical or MAC addresses. A MAC address is the one which uniquely defines one NIC, as described in Chapter 2 (Section 2.04).

The reason that unique addresses have so far been guaranteed is that the 48 bits used for the definition have given a sufficient number of different addresses. However, there is a 64-bit alternative which is already used occasionally but is available for future use when 48 bits are insufficient. The 48-bit address is usually written in hexadecimal notation, for example:

4A:30:12:24:1A:10

In one version of this addressing scheme the first three bytes identify the manufacturer.

17.06 Application-layer protocols associated with TCP/IP

There are very many application-layer protocols. This discussion considers some that are used most often.

HTTP (HyperText Transfer Protocol)



TIP

Be careful not to confuse HTTP and HTML.

Because HTTP (HyperText Transfer Protocol) underpins the World Wide Web it has to be considered to be the most important application-layer protocol. Every time a user accesses a website using a browser, HTTP is used but its functionality is hidden from view.

HTTP is a transaction-oriented, client–server protocol. The transaction involves the client sending a 'request' message and the server sending back a 'response' message. The HTTP protocol defines the format of the message. The first line of a request message is the 'request line'. Optionally this can be followed by header lines. All of this uses ASCII coding. The format of the request line is:

```
<Method> <URL> <Version>CRLF
```

where CR and LF are the ASCII carriage return and line feed characters. The request line usually has GET as the method. However, there are several alternatives to the GET method which makes HTTP potentially a more widely applicable protocol than just being used for webpage access. The version has to be specified because HTTP has evolved so there is more than one version in use.

Email protocols

The traditional method of sending and receiving emails is illustrated in Figure 17.04 which shows that three, individual, client–server interactions are involved. The email sender acting as a client has a connection to a mail server. This server then has to function as a client in the transmission to the mail server used by the email receiver acting as a client.

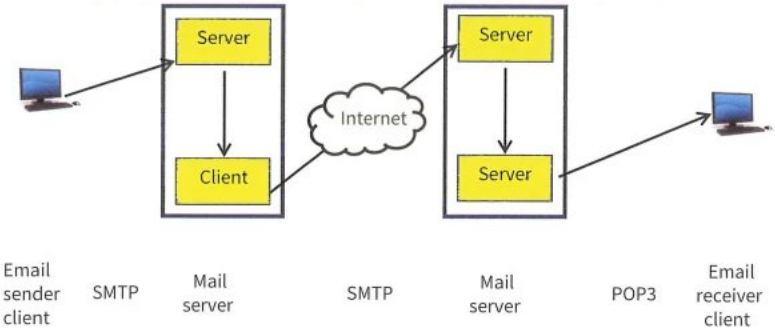


Figure 17.04 An email message being transmitted from a sender to a receiver

Of the two protocols shown being used, SMTP (Simple Mail Transfer Protocol) is a 'push' protocol whereas POP3 (Post Office Protocol version 3) is a 'pull' protocol. There is a more recent alternative to POP3, which is IMAP (Internet Message Access Protocol). The approach using POP3 is for emails to be downloaded onto the client computer. With IMAP the emails are not downloaded; they remain stored on the server but remain accessible from the client. It can be argued that POP3 is more secure to cyber-attack because emails are locally stored. However, the server will be backed up regularly whereas the local client might not be. The major advantage for IMAP is that the server can be accessed from any client. This makes it suitable for anyone on the move or if someone is using a system other than that normally used. POP3 emails are only accessible from the one client system.

SMTP has been largely replaced by the use of web-based mail. A browser is used to access the email application, so HTTP is now the protocol used. However, SMTP remains in use for transfer between the mail servers.

Question 17.02

Which protocols are being used in your email system? Does your email system give you any choices?

FTP (File Transfer Protocol)

For routine transfers of files from one user to another the most likely method is to attach the file to an email. However, this is not always a suitable method. FTP (File Transfer Protocol) is the application-layer protocol that can handle any file transfer between two end-systems. File transfer is not simple if the end-systems have different operating systems with different file systems. FTP handles this by separating the control process from the data-transfer process.

Question 17.03

If you were setting up a website, you would need to upload files to a server. Do you know how you can use FTP for this from the system you normally use?

17.07 Peer-to-peer (P2P) file sharing

The network traffic generated by peer-to-peer (P2P) file sharing is one of the main features of Internet use. P2P is an architecture that has no structure and no controlling mechanism. Peers act as both clients and servers and each peer is just one end-system. When a peer acts as a server it is called a 'seed'.

The BitTorrent protocol is the most used protocol because it allows fast sharing of files. There are three basic problems to solve if end-systems are using BitTorrent.

1 How does a peer find others that have the wanted content?

   Every content provider should provide a content description, called a torrent, which is a file that contains the name of the tracker (a server that leads peers to the content) and a list of the chunks that make up the content. The torrent file is at least three orders of magnitude smaller than the content so can be transferred quickly. The tracker is a server that maintains a list of all the other peers (the 'swarm') actively downloading and uploading the content.

2 How do peers replicate content to provide high-speed downloads for everyone?

   Peers download and upload chunks at the same time, but peers have to exchange lists of chunks and aim to download rare chunks for preference. Each time a rare chunk is downloaded it automatically becomes less rare!

3 How do peers encourage other peers to provide content rather just using the protocol to download for themselves?

   This requires dealing with the free-riders or 'leechers' who only download. The solution is for a peer to initially randomly try other peers but then to only continue to upload to those peers that provide regular downloads. If a peer is not downloading or only downloading slowly, the peer will eventually be isolated or 'choked'.

It is worth noting that the language of BitTorrent is quite unusual and there are other terms used which have not been mentioned. Fortunately, the principles are straightforward.

Question 17.04

Who runs the tracker server? Are there alternative approaches for BitTorrent?

Reflection Point:

Networking is a large subject area with many interconnected concepts. Have you considered how you might make some structured notes to help with revision later?

Summary

- Circuit switching requires a dedicated circuit to be established between sender and receiver before transmission can start.
- In packet switching, packets can be transmitted without any circuit being created.
- Packet switching can use connectionless or connection-oriented transmission.
- A protocol suite is implemented as a layered stack.
- The TCP/IP protocol suite supports usage of the Internet.
- Ethernet is the most likely option for use in the Network and Data link layers.
- Examples of application-layer protocols are HTTP, SMTP, POP3, IMAP and FTP.
- Peer-to-peer file sharing on the Internet uses the BitTorrent protocol.